

Hello everyone, I am Wenje Ma. The title of my talk is "Bayesian Optimization Based on Multi-Fidelity Data."

Let us first look at the formal problem statement. The design parameters $\mathbf{x}$ live in the p -dimensional unit interval, and the simulator is a black box y equals $h(\mathbf{x})$. Two evaluation channels are available: the low-fidelity $h_L(\mathbf{x})$ is cheap, cost c_L , but has a systematic bias; the high-fidelity $h_H(\mathbf{x})$ is expensive, cost c_H much greater than c_L , and is accurate. The bias structure between the two fidelities is unknown. The total budget is 15 high-fidelity equivalents. Define $\hat{\mathbf{x}}$ as the algorithm's best estimate; $\hat{h}_H$ equals $h_H(\hat{\mathbf{x}})$, the high-fidelity value measured there. Our goal is to estimate as accurately as possible the $\mathbf{x}$ minimizing $h_H(\mathbf{x})$.

Let us lay the theoretical foundation. We estimate x through Bayes' formula, with a prior and a posterior. At the prior stage we know nothing about the function, so the initial points must cover the space as uniformly as possible. To ensure that in every one-dimensional projection all coordinates are distinct, Morris and Mitchell proposed the maximin Latin hypercube sampling design in 1995; as Figure 1 shows, it covers the space uniformly.

However, it projects poorly in high dimensions. Joseph et al. therefore proposed the maximum projection design in 2015: for a design x_1 to x_n on the p -dimensional unit interval, the criterion is (1). Since the denominator cannot be zero, distinct coordinates are guaranteed automatically, satisfying the maximin constraint.

As Figure 2 shows, its low-dimensional projections are also uniform. We therefore adopt maximum projection for choosing the initial points.

Ranking factors by importance lets us drop the unimportant ones —fewer factors, easier fitting. The importance of factor i is the total Sobol' index V_i total, estimated by Jansen in 1999 as in (2). Since the estimator is random, we examine its expectation, giving (3).

Because τ times the square root of ω_i is constant for fixed i , maximizing (3) reduces to maximizing the L_1 distance, achieved through (4); this is the maximum one-factor-at-a-time design.

Inspired by maximum projection, Joseph in 2026 proposed letting A and B alternately apply the mapping in (5) to improve the projection properties; the two branches in (5) are interchangeable. The result is the maximum one-factor-at-a-time design with projection.

Figure 3 compares it with the version without projection —the mapped points no longer duplicate coordinates.

We do not need to spend the whole high-fidelity budget at once. We can build a small initial design, collect data, and then decide where the next experiments go. This flexible strategy is the sequential design, also called active learning in machine learning; unlike the maximin Latin hypercube design, the maximum projection design supports it naturally. Define X as the design space in the p -dimensional space of $\mathbb{R}$. Joseph proposed in 2016 choosing x_1 as the initial point, then adding x_2 to x_n one by one via (6), with k from 1 to n minus 1. In fact, (6) is just the maximum projection criterion (1) rewritten iteratively. We will use it for sequential point selection.

Suppose there are K fidelity levels, 1 the lowest and K the highest, with design sets D_i , i from 1 to K . Kennedy and O'Hagan fused the data as in (7), assuming Gaussian processes as in (8).

Recall: a Gaussian process is a random function whose values at any finite set of points follow a multivariate normal distribution —a generalization of the multivariate normal to arbitrary dimensions.

They also proposed a nested design: D 1 contains D 2, and so on down to D K. Consider the two maximum-entropy design problems in (9). Entropy, defined by Shannon in 1948, measures uncertainty; experimental design seeks larger entropy, i.e., more information —exactly what maximum projection maximizes, making it ideal for D 1 star and D 2 star.

Joseph in 2026 noted that the nested design is built by successively removing points from D_1 star to form D_2 star, with the removal criterion in (10). Unlike (6), the denominator adds a positive constant β_1 , because D_2 takes discrete values from the continuous D_1 : two points coincide with probability zero in a continuous space, but not in a discrete one, so the constant prevents a zero denominator. Figure 4 shows the nested design with two fidelity levels: the triangular D_2 points are discrete points chosen from the circular D_1 points, both following maximum projection.

Now to the core optimization problem: minimize the expensive black-box function $h(x)$ over the design space, taken as the p -dimensional unit interval, giving (11) —a nonlinear problem with box constraints. We need a global method using as few evaluations as possible. We select a small design x_1 to x_n , evaluate $h(x_i)$ to get y_i , and approximate the function by a cheap Gaussian process surrogate $\hat{h}_n(x)$, which approaches the high-fidelity surface sequentially. Minimizing the surrogate yields the next point via (12); iterating until convergence gives Bayesian optimization. The standard choice is expected improvement, proposed by Jones et al. in 1998.

Given the design $\mathbf{D}_n$ and observations $\mathbf{y}_n$, we place the Gaussian process prior (13).

By Lemma 2.2 of Joseph in 2026, the posterior is (14). Let h_n^{minimum} be the minimum of y_n and define the improvement I in (15): improvement occurs when h_x falls below h_n^{minimum} .

Since I is random, we maximize its expectation, giving the iterative rule (16). Jones et al. derived its explicit form (17).

And differentiating yields (18): expected improvement decreases with $\hat{h}_n(x)$ and increases with $s_n(x)$, balancing optimization against surrogate accuracy.

Figure 5 shows two iterations on a test function, gradually locating its minimum and second minimum.

We now pick test functions of 1, 2, 4, and 8 dimensions to compare the pipeline across dimensions. These are: the one-dimensional test function of Joseph in 2026, it's in (19); the Branin function of Surjanovic and Bingham in 2013, it's in (20);

a four-factor additive function of Joseph in 2026, it's in (21); and the borehole water flow function of Morris et al. in 1993, it's in (22).

For each test function we construct a low-fidelity version. Inspired by Song et al. in 2024, I use the general form (23): the high-fidelity function plus a constant c times the average shift from 0.5, with c set to 10 to 15 percent of the range of f_H .

The resulting low-fidelity versions are given in (24) to (27).

Now the experiments. The complete pipeline is M1 in (28); the blank control M0 in (29) drops both the sequential and nested designs.

To attribute each component's contribution, we also build S1, removing only the nested design (30), and S2, removing only the sequential design (31).

The calibration experiment fixes the budget split. We sweep the initial number of high-fidelity evaluations from 2 to 15; the value 1 is unsolvable, since a single point cannot fit a Gaussian process. For each setting we run M1 twenty times, record f star once the budget is exhausted, and take the median as the metric, choosing the n star with the smallest median. Output 1 shows the optimum: 2 for dimension 1, and 14 for dimensions 2, 4, and 8. Low-dimensional functions are easy to fit; the difficulty explodes in high dimensions. And since the optima for 2, 4, and 8 dimensions sit on the grid endpoint 14, under the tiny budget of 15 high-fidelity equivalents the initial design dominates — the sequential and fusion components have almost no room left.

The ablation experiment identifies each component’s marginal contribution while controlling the initial design. Using n 2 star directly would leave too few low-fidelity samples, degenerate the pipeline into pure high fidelity, and make the four configurations indistinguishable. So we force exactly 5 high-fidelity evaluations per dimension, repeat twenty times, and obtain Output 2.